INCH-POUND

MIL-DTL-32630/2
3 July 2019

DETAIL SPECIFICATION SHEET

CONDUCTOR, ELECTRICAL, STRANDED, UNINSULATED CARBON-BASED CONDUCTIVE FIBER, 0.010 NOMINAL DIAMETER, EXCLUSIVE, -65 DEGREES CELSIUS to +150 DEGREES CELSIUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32630.

REQUIREMENTS:

TABLE I. Conductor properties.

Part or Identifying Number (PIN)	Conductor type (see table II)	Coating type (see 1/)	Tow (no. of filaments)	Filament diameter (in.)	Min. average break load (lb.) (see 2/)	Max. DC resistance (Ω/ft.)	Max. linear density (lb./1000 ft)	Fiber CSA (in^2) (see 3/)
M32630/02-001	I	NC	89	0.00075	5	2.0	0.091	7.90E-05
M32630/02-002	II	HNC	166	0.00063	15	1.2	0.119	6.33E-05
M32630/02-003	III	HNC	80	0.00120	19	0.8	0.170	9.52E-05
M32630/02-004	I	SC	89	0.00076	5	2.0	0.093	8.00E-05
M32630/02-005	II	SC	166	0.00063	15	1.2	0.119	6.33E-05
M32630/02-006	III	SC	80	0.00120	19	0.8	0.170	9.52E-05

1/ Silver Coating (SC), Nickel Coating (NC), Heavy Nickel Coating (NHC).
2/ Minimum elongation shall be 1 percent.
3/ Cross-sectional Area (CSA) of entire fiber.

TABLE II. Conductor type.

Type	Fiber material
I	Metal Coated Aramid/para-aramid Fiber (MCAF)
II	Metal Coated PBO Fiber (MCPF)
III	Metal Coated LCP Fiber (MCLF)

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1. The tensile strength of conductors manufactured to this specification sheet shall be in accordance with table III.

TABLE III. Tensile strength.

Conductor type	Minimum tensile strength (KSI)
I	127
II	250
III	200

TABLE IV. Size 26 conductor types.

Part or Identifying Number (PIN)	Conductor diameter (in) (±.001)	Max. DC resistance Size 26 conductor (Ω/ft.)	Min. average break load (lbs.)	Min. fusing time (sec.) (at 3.25A)	Rated current (Amperes)	Salt fog (hrs.)	Min. flex life (cycles)
M32630/02-007	0.0180	0.67	15	90	1.25	500	500
M32630/02-008	0.0170	0.47	30	90	1.25	500	
M32630/02-009	0.0170	0.36	27	90	1.25	500	
M32630/02-010	0.0180	0.67	15	90	1.25	40	
M32630/02-011	0.0170	0.47	30	90	1.25	40	
M32630/02-012	0.0170	0.36	27	90	1.25	40	

The following requirements are specified for table IV size 26 conductors.

1. Minimum elongation of 2.5 percent.

2. Twists per inch: 2. When requested by the manufacturer, the qualifying activity can approve use of different values.

3. Balance ply: Required.

4. Temperature coefficient of resistance: The temperature coefficient of resistance for size 26 conductors shall be in accordance with table V.

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TABLE V. Temperature coefficient of resistance for size 26 conductor.

Part or Identifying Number (PIN)	Temperature coefficient of resistance ($\Delta\Omega/\Omega$@20˚C)
M32630/02-007 Type I	0.003966
M32630/02-008 Type II	0.00421
M32630/02-009 Type III	0.00401
M32630/02-010 Type I	0.003966
M32630/02-011 Type II	0.00386
M32630/02-012 Type III	0.00442

NOTES:

1. Diameter (as-round): 0.008 - 0.012 inch when twisted to round under 0.5 lb. tension

2. Flat Width (as-braided):

TABLE VI. Flat width.

Conductor type	Nominal flat width (in.)
I	0.020
II	0.027
III	0.024

3. Thickness (as-braided):

TABLE VII. Fiber thickness.

Conductor type	Nominal fiber thickness (in.)
I	0.004
II	0.007
III	0.004

4. Nominal metal percentage: 75 percent of the fiber weight

5. Typical weight savings compared to copper: 75 percent

6. Intended use. This conductor is intended for use in EMI shield and multi-strand conductor applications.

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7. The conductor is manufactured in bulk lengths (ten thousand to hundreds of thousands of feet per lot) and supplied on bobbins in lengths for further processing (hundreds to thousands of feet per bobbin) in braiding, weaving or multi-stranding operations.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS
Project 6145-2018-052

Review activities:
Army - AR
Navy - SH
Air Force - 11, 20

NOTE: The activities above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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